

1 UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

2 DOCTORAL THESIS

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4 **The study of Higgs properties in the**
5 **four-lepton final state at CMS**

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8 *A thesis submitted in fulfillment of the requirements*
9 *for the degree of Doctor of Science*

10 *in the*

11 **Institute of High Energy Physics**
12 **Chinese Academy of Sciences**



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Abstract

Institute of High Energy Physics
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The study of Higgs properties in the four-lepton final state at CMS

by Tahir JAVAID

0.1 Introduction

The measurement of properties of Higgs boson is presented in its four-lepton decay channel. The fiducial cross sections (integrated and differential) of the Higgs boson production via $H \rightarrow 4\ell$ decays ($\ell = e, \mu$) are measured using the data collected with CMS detector in pp collisions at $\sqrt{s} = 13$ TeV . The measurements use Higgs mass $m_H = 125.09$ GeV and correspond to an integrated luminosity of 137.1 fb^{-1} at 13 TeV (Full Run 2 data) from the Large Hadron Collider (LHC) experiment. The differential fiducial cross sections are presented as a function of several kinematic observables which are sensitive to the Higgs-boson production mechanisms. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

0.1.1 Motivation

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) using the full Run 2 CMS data recorded in proton-proton collisions at $\sqrt{s} = 13$ TeV are presented. Included Higgs differential measurements are based on Higgs kinematics and motivated from several aspects. For instance, Higgs trasverse momentum can be used to test the pQCD calculations (especially when measured with exclusive multiplicities of jets). Additionally this variable is sensitive to Lagrangian structure of Higgs. Rapidity of Higgs is sensitive to parton distribution functions (PDFs) of the protons involved in the collisions. Jet related observables which are sensitive to theoretical modelling, boosted quark and also to the gluon emission are also covered in the dissertation. Current measurements are consistent with SM prediciton; however, in such measurements, any deviation from SM expectations provides a direct hint to BSM contributions. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different decay cahannels of Higgs which can been studied in Ref. [Khachatryan:2015yvw, CMS:2015hja, CMS:2016rqf, Sirunyan:2017exp, CMS:2018hhg, ATLAS-CONF-2019-029, atlas-conf-2019-025, ATLAS:2018uso, ATLAS-CONF-2019-029, ATLAS:2017myr, CMS:2019chr, Sirunyan:2020xwk, 1, 2].

0.2 Theoretical Background

The measurements are motivated from the theretical predictions from well-known Standard Model (SM) of particle physics. After a long period of research, ordinary matter and underlying forces or interactions (electromagnetic, weak and strong interactions and not the gravity) which bind it in a structure are successfully explained by the SM. Details on the fundamental forces are described in Section ???. In fact, Higgs boson is manifestation of arguments from core components of Standard Model which was first discovered at CERN,LHC facility in proton proton collisions. In SM, description of these interactions is made by symmetry laws of mathematics which could only explain the characteristic of particles with zero mass. Higgs mechanism was then introduced to make the theory realistic enough to incorporate mass problem. Under this mechanism, electroweak symmetry breaks and as a result particles gain the mass. Higgs boson is the direct manifestation of this Higgs mechanism [3, 4, 5, 6, 7, 8].

SM is a quantum field theory (QFT) belonging to $SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma$ gauge symmetry group. It was formulated from a collaborative work spanned throughout the 20th century. In 1961, there was a proposal from Glashow that electromagnetic and weak intractions are similar substructures of electroweak intraction and presented their unified interpretation with $SU(2)_L \otimes U(1)_\gamma$ symmetry group. Brout, Englert, Higgs *et al.* proposed a mechanism with which fundamental particles may acquire mass through the idea of spontaneous symmetry breaking and associated

new elementary particle called Higgs boson. Salam and Weinberg then annexed this idea to the Glashow's model $(SU(2)_L \otimes U(1)_Y)$, where " L " represents left handed chiral states of fermionic fields. " Y " is the weak hyper charge and can be related to the weak-isospin (T) and electric charge (Q). This completed the EW sector of SM. Glashow's proposal of unification of EM and weak forces described the kinematics of massless particle only as described in Section ?? . However, gauge fields must acquire mass in order to correctly describe weak interaction and to be consistent with experimental findings. The Higgs mechanism was postulated as a part of SM, which introduces the existence of a new scalar Higgs field. This field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of masses of the fermions and bosons ($W^\pm$, Z boson) are introduced [3, 4, 5, 6, 7, 8]. Details to the Higgs mechanism are described in section ??.

0.2.1 Phenomenology of proton-proton collisions at LHC

LHC is the largest colliding experiment of its kind ever built also motivated to perform test of the SM of particle physics. Protons are composite particles made up of three valence quarks (uud) which are bound together with strong force exchanging the gluons. There are several ways through which they can interact within a proton. Quarks can mutually interact with other quarks or through annihilation process they can produce gluons which can further go for pair-production of quark anti-quark pair. As a results, apart from the valence quarks, there is a sea of virtual quarks and gluons inside a proton (collectively called partons). In a collision experiment, it may happen that there is large momentum transfer from the colliding partons. This is called hard scattering and as a results many resonances can be produced e.g. Higgs. Probability of a certain process to occur is known as cross section of that process. The production cross sections of different process at LHC in the proton-proton collisions are shown in the Figure. ?? in next chapter.

Partons which do not take part in hard interaction, usually go straight and may involve in certain low momentum transfer processes. These activities of low momentum transfer are called underlying events. In case if more than one interactions happen in a certain proton proton collision, it is referred as multi-parton interactions (MPI).

In an accelerated environment, charged partons may radiate photons or gluons through a process known as "Bremsstrahlung". If this radiation is associated with the incoming partons, it is called initial state radiation (ISR) and if it is coming from outgoing parton from the interaction point, it is called final state radiation (FSR). The quarks and/or gluons when they go through the process of hadronization (bind together to form color singlet) make a bunch of hadrons usually in the form of a cone which are called "jets" and are described in Section. ??.

In hard scattering experiments, probability of a certain process to occur is known as cross section and can be calculated using the Factorization theorem. This theorem helps to disintegrate the cross section into hard and soft processes using the

scale which is known as factorization scale (μ_F).

Mathematical description for the cross section is given in section ??.

0.2.2 Higgs production and decay

There are five main production modes of Higgs are included in the analysis upon which this dissertation is built. The most dominant production mode of Higgs at LHC is gluon-gluon fusion. The gluons couples to Higgs through the loop of virtual quarks, as coupling is directly proportional to fermion mass, top quark loop is most dominant. VBF is second most important production mode of Higgs at LHC and as well as LEP. In VBF production mode, two $W^\pm$ or Z boson fusion create Higgs that have been emitted by a (anti)fermion pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in detector in its forward region. These jets signature helps to increase signal to background ratio. In W/Z association production mode, Higgs is radiated by high energetic $W^\pm$ or Z bosons that is produced by fusion of two fermions. This production mode was leading mechanism at LEP and sub-leading mechanism at Tevatron. At LEP it becomes very probable to produce $W^\pm$ or Z boson with electron positron collisions. In case of hadron collider, Tevatron makes quark antiquark collisions more likely as compared to LHC. The least dominant process of Higgs production is illustrated in Figure ?? (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association $t\bar{t}$.

The life time of SM 126 GeV Higgs is 1.6×10^{-22} s and it can decay to all SM particles having masses (most probably to heaviest onshell particle). Higgs boson decay branching ratio to a variety of particles are illustrated in Figure ?? [9]. $H \rightarrow b\bar{b}$ (at $m_H \approx 125$ GeV) is the most dominant decay mode of Higgs. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, CMS and ATLAS collaborations and other collaborations [10], [11] and [12](CDF and D0) have exploited this channel, in association with a vector boson which decays to leptons, in boosted regime. B quarks are also difficult to reconstruct in a detector. When they decay in the detector, they fragment into jets of many hadrons. These jets are difficult to reconstruct perfectly. Because of this, it is necessary to exploit other channels in this region. $H \rightarrow \gamma\gamma$ is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution.

As the Higgs mass at 125 GeV, Higgs boson decay to diboson(WW, ZZ) boson are also available, where one or both boson can be off-shell. WW decay further has neutrinos in final state due to which it is not possible to reconstruct the final state fully. $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can be reconstructed with excellence in the detector [9]. Measurements on Higgs reported in this dissertation are based on $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ channel.

0.3 Experimental Background

Precision measurements on the high energy physics experiments like Tevatron and LEP have proven the authenticity of SM; however, Higgs observation could not happen. In 2010, Large Hadron Collider (LHC), the largest experiment ever of its kind in physics world, was then set to put more tests on SM and it endorsed the measurements on SM from earlier experiments. LHC is located at CERN (European Organization for Nuclear Research) at Franco-Swiss border near Geneva, Switzerland and is most powerful setup of its kind to probe resarches in particle physics. From its design, it is capable enough to accelerate and then collide two proton beams at center of mass energy upto 14 TeV at peak luminosity of $\mathcal{L} = 10^{34} \text{cm}^2 \text{s}^{-1}$. It is the most advanced, complex and largest experiment in the history of mankind, designed and constructed by a collaborative team of more than 10,000 technician, engineers and scientists from over 100 countries. It was constructed from 1998 to 2008, in a 3.8 meters wide tunnel of 27 Kilometers circumference at a depth which varies from 75-150 meters underground. This tunnel was previously used for LEP collider which was constructed from 1983 to 1988. Two counter rotating proton beams (anti-proton beam is disfavored to use) in two rings are used where the beam pipes are kept at ultra-high vacuum just to preserve the beams from possible extra collisions with molecules of gas. In 2012, the “hidden” Higgs boson, was discovered in shape of heavy scalar boson as jointly reported by the CMS and ATLAS experiments of LHC(CERN) [13, 14]. With this happening, physicists were all set to measure other properties of Higgs for further tests on SM predictions. Mass of Higgs boson was only free parameter in Higgs sector of SM until measured to be $m_H = 125.09 \pm 0.21(\text{stat.}) \pm 0.11(\text{syst.}) \text{ GeV}$ [15].

After successful data taking during Run 1, LHC machine was shutdown for two years in 2013 and 2014 for major upgrades in LHC machine and detectors. LHC started again in 2015 at center-of-mass energies to 13 TeV(close to its design value i.e 14 TeV). LHC delivered 4.2 fb^{-1} during 2015 at 13 TeV center of mass energy and subsequently, it delivered 41.0 fb^{-1} , 49.8 fb^{-1} and 67.9 fb^{-1} during the years 2016, 2017 and 2018 as shown in the Figure ?? . After Run2 operations, LHC is passing through its long shutdown 2 period. During this period, several upgardees are ongoing which can be regarded as preparations for High Luminosity LHC later on (HL-LHC). However, main upgrades in this regards will take place during LS3 after Run 3 period of LHC. One is the replacement of Linac2 with the Linac4. It is about 90 meter in length that took around 10 years to build it. It would be capable to accelerate negative hydrogen ions (H^-) for beam energy upto 160 MeV and will enable beam to double its intensity Ref. ?? . Besides, the individual experiments of LHC, i.e. CMS, ATLAS, ALICE and LHCb, have also planned certain upgarades during this LS2 period Ref. ?? .

ATLAS is builing muon small wheels to be integrated with the trigger system. For high pileup LAr electronics are being upgraded.

CMS is changing the photodetectors (HPD) of barrel and endcap HCAL to the silicon photo-multipliers (SiPM). Also the beam pipes made with stainless steel are being replaced with Aluminium composed material. Also, based on damage from

radiation in endcaps of HCAL, scintillating tiles would be changed. Furthermore, for HL-LHC preparations, CSC muon chambers are being upgraded with new front end (FE) electronics.

ALICE is making several upgrades for Inner Tracking System (ITS), Time Projection Chamber (TPS), Central Trigger Processor, Data Acquisition (DAQ) High Level Trigger (HLT), Muon Forward Tracker (MFT) and so on.

LHCb is improving its readout from 1 MHz to 40 MHz with trigger based fully on softwares by replacing FE electronics and new computing infrastructure.

0.3.1 The Compact Muon Solenoid detector

The data used for the analysis in the dissertation was collected using the CMS detector. More details on the geometry of CMS, its parts (sub-detectors) and description of how events are reconstructed with the CMS detector can be seen in Chapter ?? . LHC has set new frontiers for researches in particle physics experiments. State of the art experimental facility has annexed with equipment based on latest technology. To assure the validity of results from the experiment, LHC has its two major and general purpose experiments for cross-validation, one of which is CMS. They have the same physics goals with some different design and the detector technology. Particularly the CMS is capable to provide good resolution for muon identification and momentum over wide range of dimensions and determination of the charge on muons with ultimate precision with momentum up to 1 TeV. It gives healthy inner tacker reconstruction efficiency and resolution of charged particle momentum. It has efficient trigger system and better discrimination of τ 's and b-jets. Further, it is benefitted from good dielectron, diphoton and dijet mass and energy resolutions, and as well as missing-transverse-energy (MET) resolutions. This makes it capable to study large areas from SM Physics (including Higgs Physics) to search for extra dimensions and the dark matter covering physics in energy range from a few 100 GeV to TeV scale [16].

Being one of the two general purpose detectors installed at LHC ring, CMS is is constructed in cylindrical shape having length about 28 m and outer radius of around 7 m. Its overall height is about 15 m with overall weight of about 14,000 tonnes. Overall design of the CMS detector can be seen in the Figure ?? . Learning from the earlier experiment at CERN, CMS engineered to be built in different slices at surface instead of doing everything underground. And fifteen separate slices were then lowered to underground to be fixed at the LHC tunnel.

Moving radially outwards from interaction point of beams, there are inner tracker system (ITS), electromagnetic calorimeter (ECAL) and hadronic calorimeter (HCAL) sub-detectors of CMS which are fitted inside the large solenoid magnet. Inner tracker of CMS have large track multiplicity as being closest to beam pipe where collisions happens. ECAL is build with crystals of lead tungstate which covers pseudorapidity up to $|\eta| < 3.0$. After ECAL, hadronic calorimeter is installed which is made of brass. Outermost layer of the CMS detector is the Muon System.

LHC produces around 40 million collisions in a second (40 MHz); however, not all the collisions are interesting for physics purposes. Further, for the data acquisition (DAQ) system, it is not effective to read every event produced at high rate. Also, storing the uninteresting events produced in LHC collisions will require a lot of storage. Hence, there is a dire need to reduce the rate of produced events during LHC collisions which is done by the trigger system which select only those events which are crucial for physics purposes. Trigger system of CMS is a sophisticated two-level system; Level 1 (L1) trigger and high level trigger (HLT).

0.4 Measurement strategy for Higgs fiducial cross section

The measurement starts with the descriptions of the SM Higgs boson production which are obtained using the POWHEG V2 generator for the five main production modes (ggH, VBF, WH, ZH and ttH) and are called signal samples. WH and ZH production modes are simulated by the MINLO HVJ extension of the POWHEG [17]. Higgs decay to four lepton is simulated by JHUGEN generator [18]. For the WH, ZH and ttH modes, Higgs is allowed to decay to $H \rightarrow ZZ \rightarrow 2\ell 2X$ such that two leptons from 4-lepton events are produced from the decay of associated W, Z bosons or top quarks respectively. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the default NNPDF 3.1 NLO parton distribution functions (PDFs) [19]. The list of signal samples with the corresponding cross sections are shown in [CMS:2019chr]. These samples, for each production mode, are also used to assess model dependence of measured results as shown in the Table ?? . The details of the method used to assess the model dependence will be described in Section ?? . The POWHEG and NNLOPS based samples for $gg \rightarrow H$ production are utilized as theoretical predictions based on SM calculations for the comparison with measured results.

For the estimations of the backgrounds, POWHEG V2 interfaced with PYTHIA8 was used to generate the dominant process $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ at the NLO accuracy level. The $gg \rightarrow ZZ$ process is simulated at leading order (LO) with MCFM. Further details on simulation and data samples are given in the sections ?? and ?? .

The measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region that matches the reconstruction level selection. For Higgs hunting, firstly four-lepton candidates are built from the selected leptons, that pass the $SIP_{3D} < 4$ (SIP_{3D} is significance of impact parameter obtained by dividing the impact parameter with associated uncertainty) vertex constraint and the isolation cuts where FSR photons are subtracted. **Z candidates** are built from oppositely charged and same flavoured leptons (e^+e^- , $\mu^+\mu^-$) with additional requirement on dilepton mass as $12 < m_{\ell\ell(\gamma)} < 120 GeV$. **ZZ candidates** are then built from non-overlapping Z boson candidates. The one who is reconstructed with $m_{\ell\ell}$ closer to

the SM Z boson mass is regarded as Z_1 and other as Z_2 . **Signal region** is built from the events containing one selected ZZ candidate at least. More details are given in section ??.

Integrated and differential fiducial cross section for $pp \rightarrow H \rightarrow 4\ell$ are measured by performing a maximum likelihood fit of the signal and background parameterisations to the observed 4ℓ mass distribution, $N_{\text{obs}}(m_{4\ell})$, and the fiducial cross section (σ_{fid}) is directly extracted from the fit. The systematic uncertainties are included in the form of nuisance parameters and are effectively integrated out in the fit procedure. The results are obtained using an asymptotic approach [20] with a test statistic based on the profile likelihood ratio [21]. This procedure for the unfolding of the detector effects from the observed distributions is the same as in Refs. [22] and [1]. In the case of the differential cross section measurements, the finite efficiencies and resolution effects are encoded in a detector response matrix which describes how events migrate from a given observable bin at the fiducial level to a given bin at the reconstruction level. This matrix is diagonally dominant, with sizeable off-diagonal elements for observables involving jets.

The dominant sources of systematic uncertainty are the experimental uncertainties in the lepton identification efficiencies and luminosity measurement, and the theoretical sources of uncertainty are found to be subdominant. In order to assess the model dependence of the measurement, the unfolding procedure is repeated using different response matrices created by varying the relative fraction of each SM production mode within its experimental constraints. The uncertainty is determined to be negligible with respect to the experimental systematic uncertainties.

0.5 Results and outlook

In order to make all the measurements independent of how Higgs is produced, all the cross sections are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})$ fb at 13 TeV. The differential fiducial cross sections are presented as a function of rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. The differential measurements are done in bins of the observables. For each kinematic or differential observable, selection procedure of the binning boundaries has been optimized in order to have a similar expected measured cross-section for every bin of the observable. For the differential cross section measurements, the background is to be subtracted at the reconstruction level for each bin of the observable. The mass spectrum for the background is not same for all differential bins because it does not have a resonance structure. For the irreducible and reducible backgrounds, the four-lepton mass spectra are obtained as the $m_{4\ell}$ template shapes through simulated events and the control regions in the data respectively for each bin of the observable for differential measurements as the procedure is described in Ref. [23].

These measurements are important to test the Standard Model predictions in the Higgs sector and can be used to constrain phase spaces of many theories beyond the Standard Model. All the integrated and differential measurements are compared with the theoretical predictions based on the Standard Model and found to be consistent within their uncertainties. More precise Higgs boson properties measurements are needed to spot any possible deviation from SM predictions which can be a doorway to new physics.

There are certain things that can be done which are beyond the scope of this dissertation. With more data, we can get more precise results. With full Run 2 data, more differential observables can be studied. Current measurements are based on POWHEG and NNLOPS prediction. However, additional theoretical prediction(s) for comparison with the observed cross section in differential bins can be used. Further, potential theory interpretations for differential observables can be studied.

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